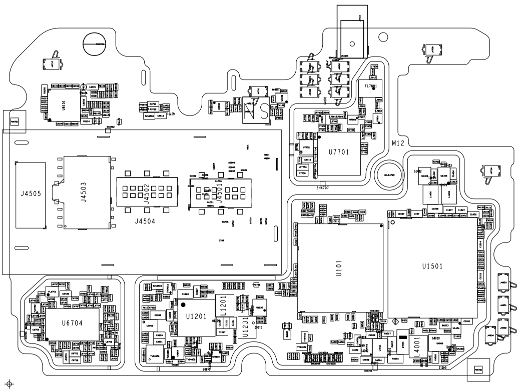
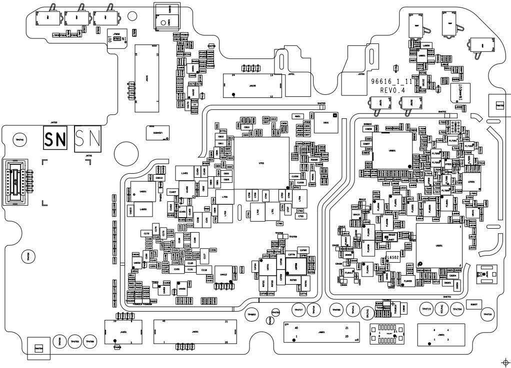
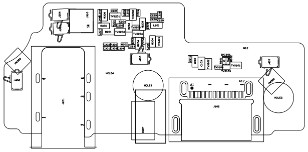
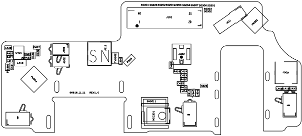


M12
FL7801
U7701
U8101
J4505
J4503
J4502
J4501
J4504
U101
U1501
U6704
U1201
L1201
U1231
L4001
SH4707


96616_1_11
REV0.4
SN
SN


M12
HOLE4
HOLE3
HOLE2
J102
A1
A12
J201
J406
J405
J407
J310
J304
R305
L201
TVS201
TVS203
TVS204
TVS101
TVS103
TVS104
EMI101


SG304 SG226 SG202 SG201 SG205 SG204 SG307 SG306 SG301
SG303
SG302
J101
40
21
1
20
J411
C428
U401
C427
R424
R421
L411
L416
J311
S N
C419
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R426
L414
L401
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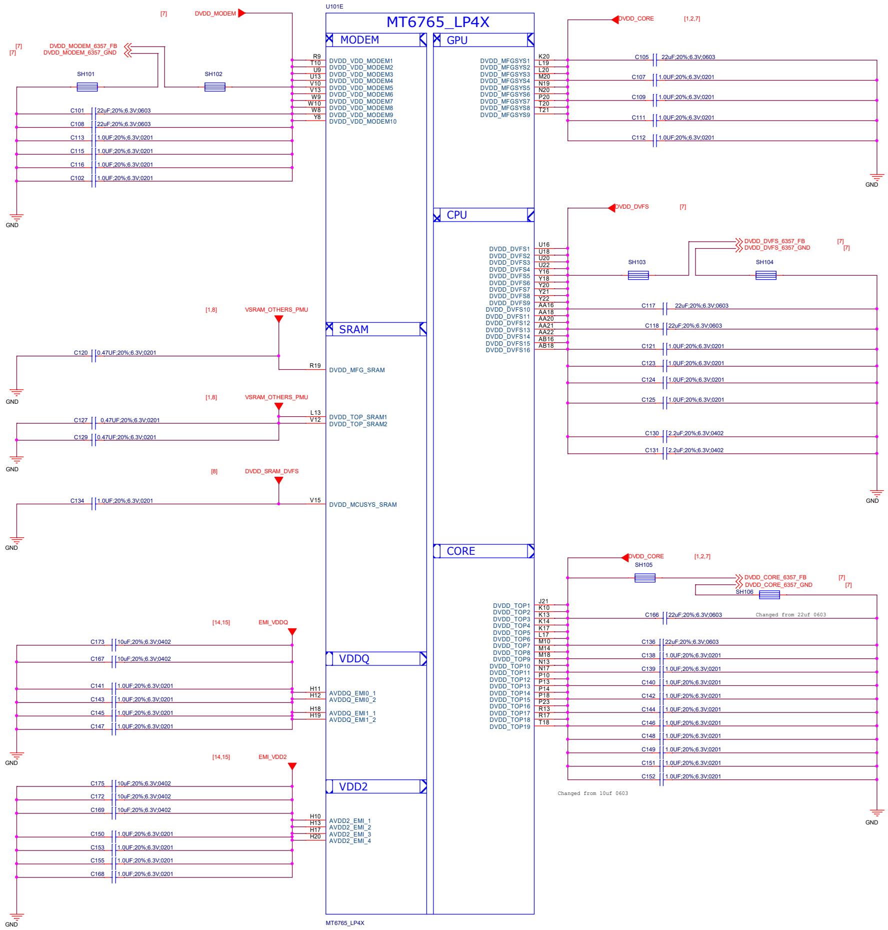
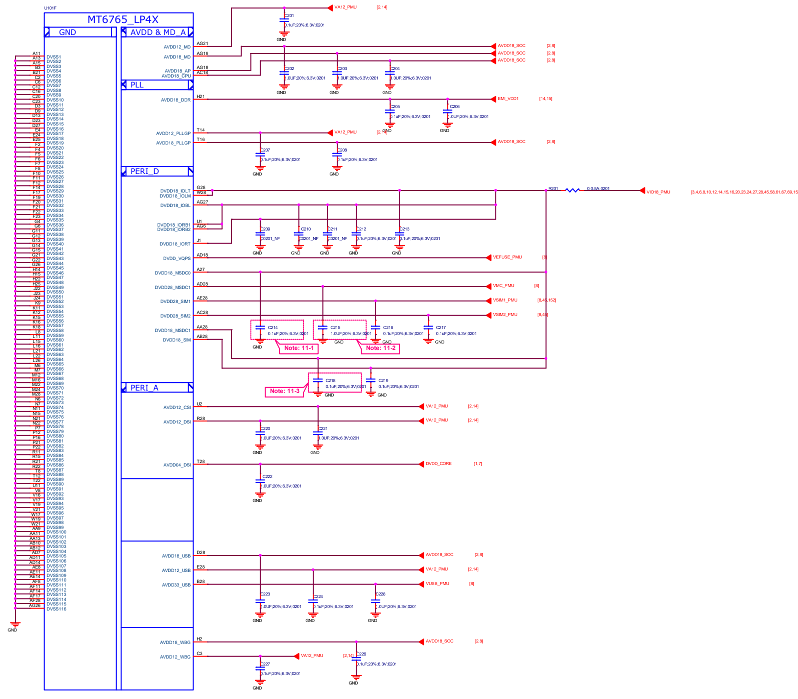
J309
C408
C430
98616_2_11 REV1.0



Title	01_BB_ POWER_PDN		REV: V10
DOCUMENT NO.:	Design Name		Size A2
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Saturday, May 21, 2022	Sheet 1	of 53



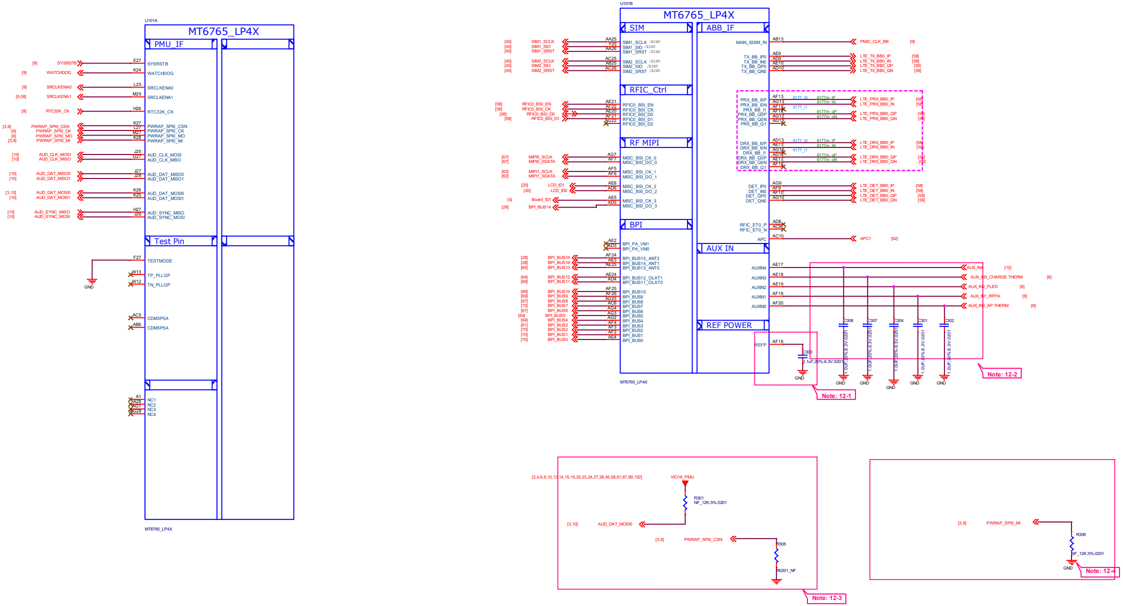
Schematic design notice of "11_BB_POWER_IO" page.

Note 11-1: C214 closed DVDD18_MSDC0 150mil

Note 11-2: C215 closed DVDD28_MSDC1 150mil

Note 11-3: C218 closed DVDD18_MSDC1 150mil

Title	02_BB_POWER_IO		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	liufenglei
WINGTECH			
Date:	Saturday, May 21, 2022	Sheet 2 of 84	



Schematic design notice of "12_BB_1" page.

Note 12-1: The de-coupling cap. for REFP (AF18 ball) have to be placed as close to BB as possible.

Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 12-3: "PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" are bootstrap pin to select which interface will be the JTAG pin out.

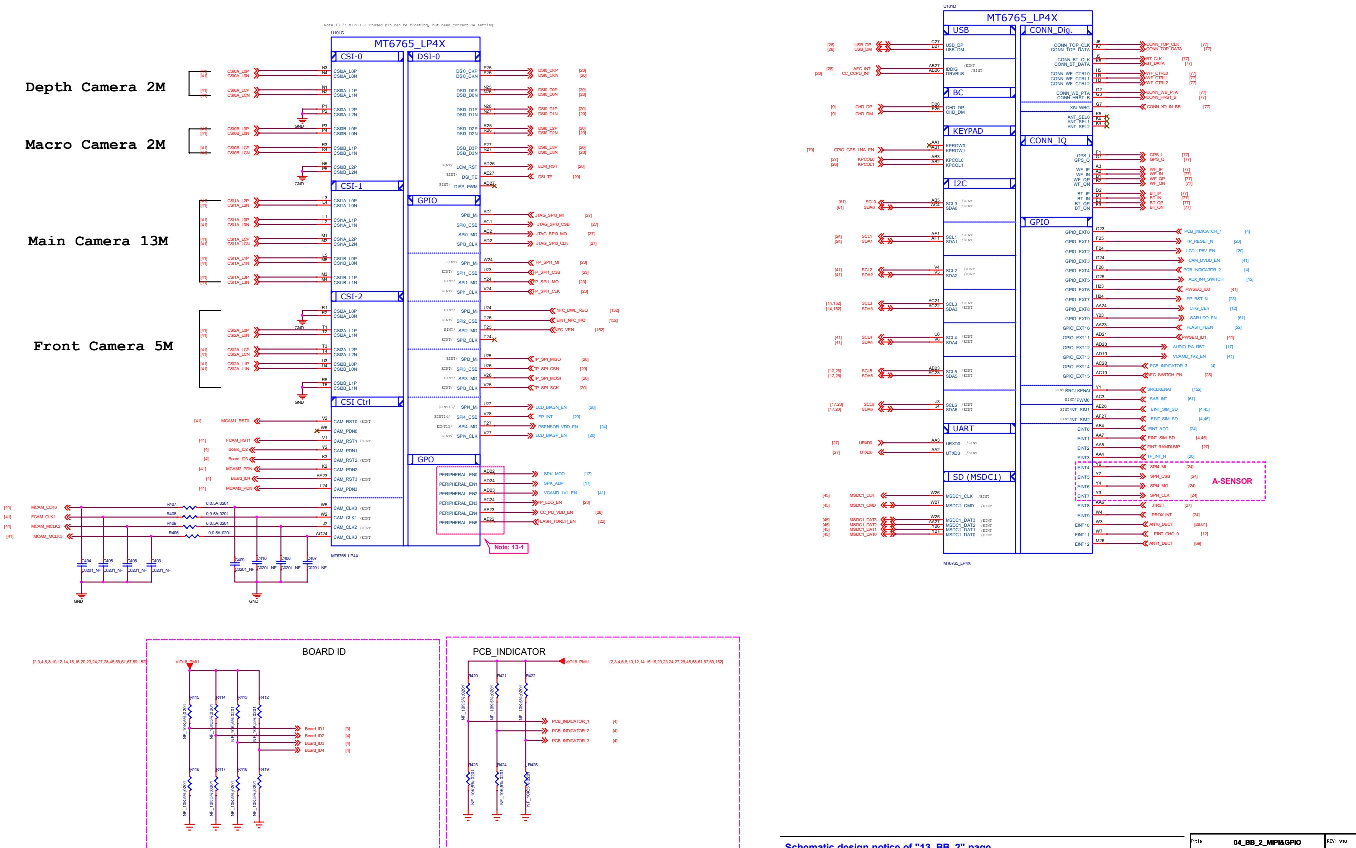
PWRAP_SPI0_CSN	AUD_DAT_MOSI0	JTAG Function	
default=PU	default=PD	AP_JTAG	MD_JTAG
HI	LO	N/A	N/A
HI	HI (by ext. PU)	SPI0+EINT8	SPI1+SPI3
LO (by ext. PD)	LO	SPI0+EINT8	N/A
LO (by ext. PD)	HI (by ext. PU)	MSDC1	N/A

Note 12-4: PWRAP_SPI0_MI / PWRAP_SPI0_MO is DDR type feature in bootstrap

PWRAP_SPI0_MI	PWRAP_SPI0_MO	Booting interface	
default=PU	default=PD	DDR	MSDC0 pin mux
LO (by ext. PD)	LO	LPDDR3	follow LPDDR3 Ref SCH.
LO (by ext. PD)	HI (by ext. PU)	N/A	N/A
HI	LO	LPDDR4X	follow LP4X/LP4 Ref SCH.
HI	HI (by ext. PU)	LPDDR4	follow LP4X/LP4 Ref SCH.

Note 12-5: Please set unused IQ pins in NC

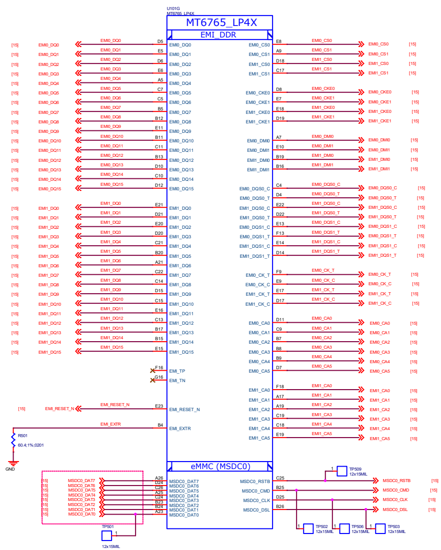
Title	03_BB_1_RF&SIM_IF	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet	3 of 46



Schematic design notice of "13_BB_2" page.

Note 13-1: The enable pin of acoustic or optoelectronic devices (e.g. SPK AMP/Backlight/Charger OCP/OVP) suggest to use Peripheral_EN[0:5]
If use other GPIOs as enable pin, suggest to reserve 0201 NC to GND

Title	04_BB_2_MIPI&GPIO		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet 4 of 46	



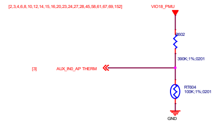
Schematic design notice of "14_BB_3" page.

Note 14-1: R501 please select 34.8 ohm (1%) resistor

Note 14-2: Please check eMCP LP3 and eMCP LP4X pin mux

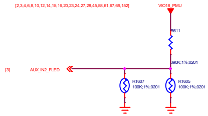
Title	05_BB_3_EMMC	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	liufenglei
Date:	Saturday, May 21, 2022	Sheet	5 of 84

The pull up ressister need to be 1% accuracy
NTC=100K

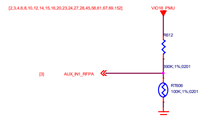


Thermistor to sense AP temperature

1. RT604must keep a distance about 6~8 mm away from AP and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.

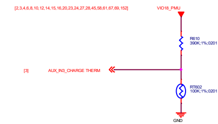


Thermistor to sense Flash LED



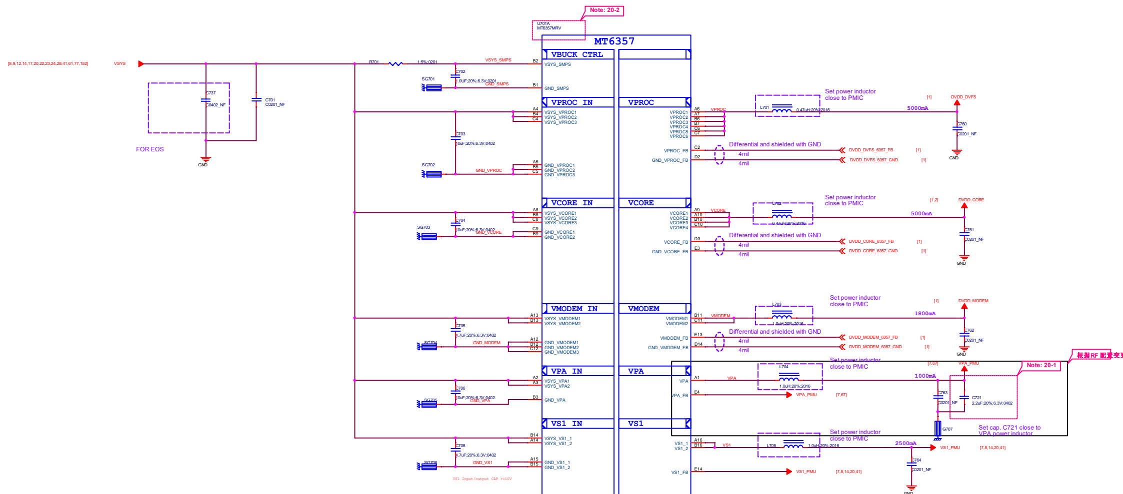
Thermistor to sense RF PA temperature

1. RT606 must close to LTE Band 7 PA or the hottest PA <2mm.
2. The distance is the shortest distance from package edge to edge.



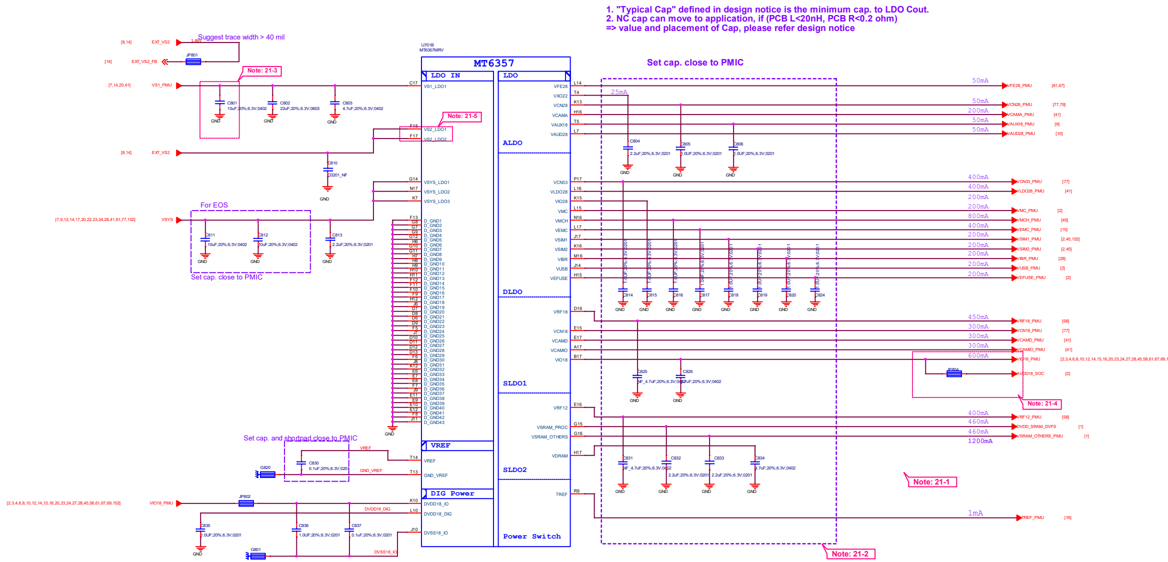
Thermistor to sense CHARGE temperature

Title	06_BB_AUXADC_Thermal	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet 6 of 46	



Schematic design notice of "20_POWER_MT6357_Buck"	
Note 20-1:	C707, please choose 0402 size
Note 20-2:	For MT6765/62/61 platform, please only use MT6357CRV

Title	07_POWER_MT6357_Buck	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
WINGTECH			
Date:	Saturday, May 21, 2022	Sheet	7 of 46



Schematic design notice of "21_POWER_MT6357_LDO"

Note 21-1: If these power trace can meet LDO layout constraint, these CAP can be NC or removed. Please refer to MT6357 design notice.

Note 21-2: Output cap range please follow MT6357CRV LDO design notice

Note 21-3: Ext Buck BOM option

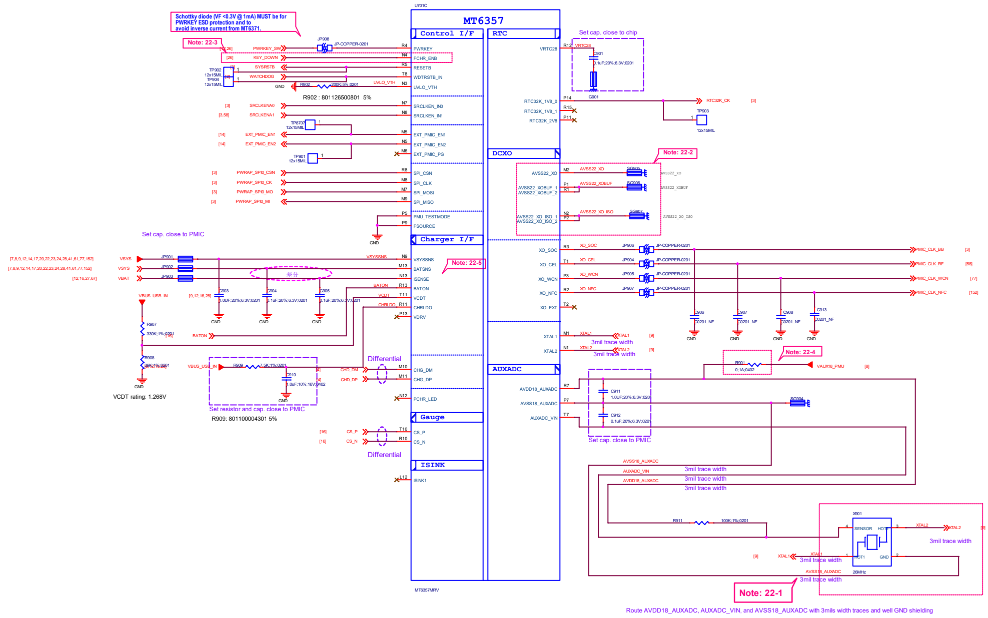
	Ext. buck option	
	w/ EXT VS2 Buck	w/o EXT VS2 Buck
C801	10uF	22uF

Note 21-4: Please set R803 and R803 close to C826, making star connection among VIO18_PMU, AVDD18_SOC, and EMI_VDD1 near to LDO cap. C826

Please also refer to MT6357 design notice for further detail design information

Note 21-5: Please connect VS2_LDO1(F15) to VS1_PMU if voltage applied to VCAMD(E17) >= 1.3 V

Title	08_POWER_MT6357_LDO	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet 8	of 46



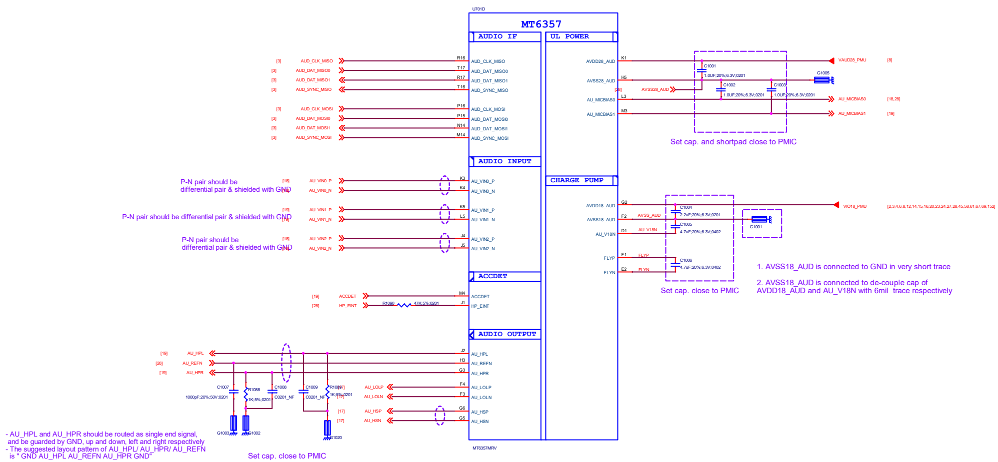
Schematic design notice of "22_POWER_MT6357-IF"

Note 22-1:	Please implement 2520 & 2016 Size TMS PCB co-layout. Please refer to MT6762_MT6357 Co-Clock Design Notice for co-layout guide
Note 22-2:	1. Please Connect P1 and R1 ball first and then to GND 2. Please Connect P2 and N2 ball first and then to GND 3. Please connect DCXO GND to main GND by independent L1-2 GND via.; DO NOT connect it through L1 GND
Note 22-3:	Let floating if disable HOMEKEY function
Note 22-4:	Please follow MT6762_MT6357 Co-Clock Design Notice for Layout guide of VAUX18, then R8101 can use 0 ohm to replace BEAD. Please route VAUX18_PMU with well-ground sheilding if using 0 ohm to replace BEAD for AVDD18_AUXADC

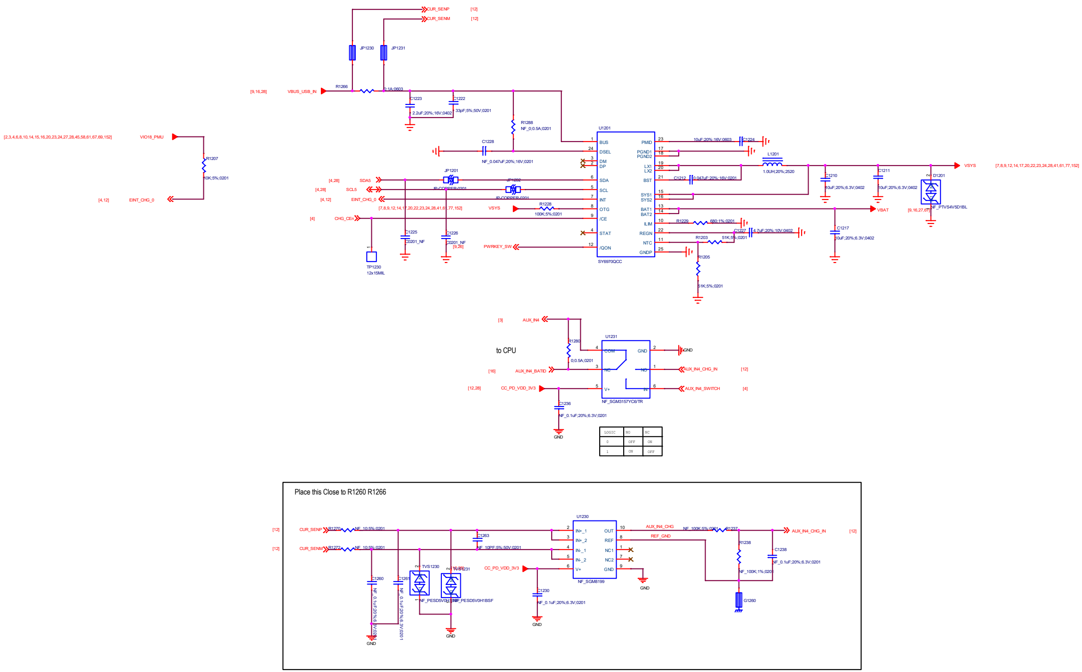
Note 22-5: Please connect to battery connector

Title	09_POWER_MT6357_IF	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet	9 of 46

WINGTECH

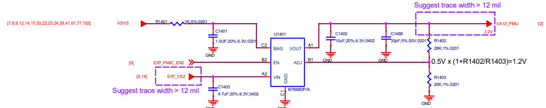


Title	10_POWER_MT6357_Audio	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet	10 of 46



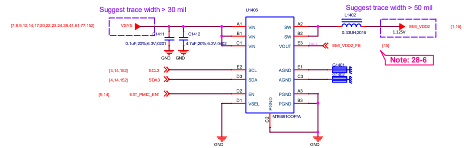
Title 12_POWER_SubPMIC-Charger + PP			REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
WINGTECH			
Date:	Saturday, May 21, 2022	Sheet 12	of 46

LDO for VA12 Note: 28-1

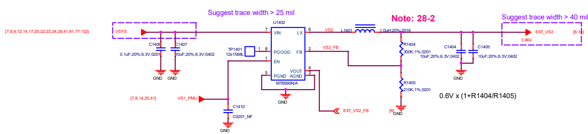


Ext. buck LPDDR4X/LPDDR4 VDRAM Note: 28-4

MT6691OOP/A / Ext. buck LP4X/LP4 VDRAM (VDD2)
I2C address: 0X57 (Write:0xAE, Read:0xAF)



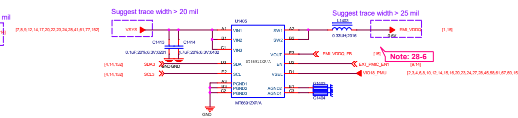
Ext. Bulk for VS2



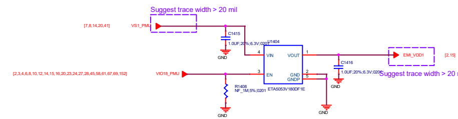
NOTE: Do not use VCAMD 1.3/1.5/1.8 when VS2 BUCK applied

Ext. buck LPDDR4X VDDQ Note: 28-4

MT6691ZXP/A / Ext. buck LP4X VDRAM(VDDQ)
I2C address: 0x50 (Write:0xA0, Read:0xA1)



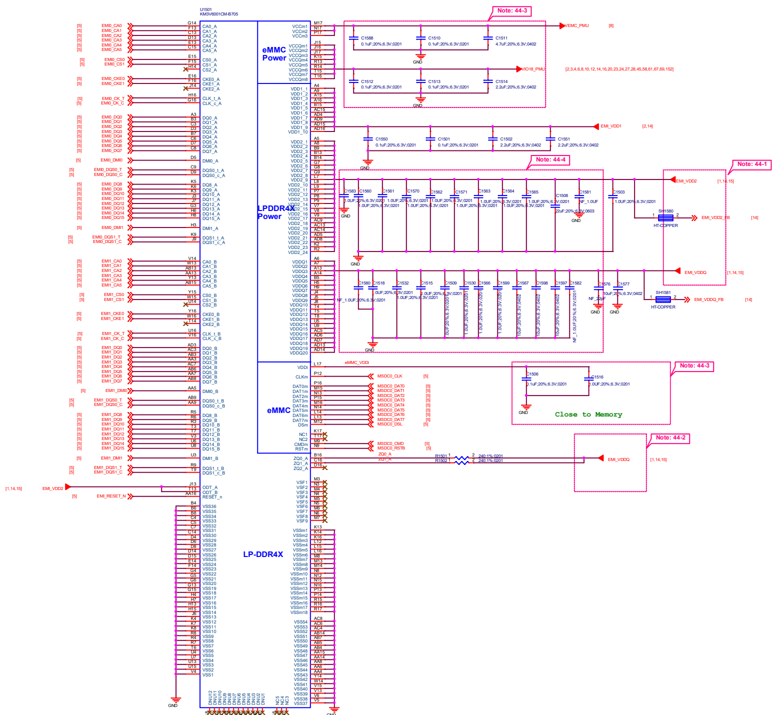
LPDDR4X/LPDDR4 VDD1 1.8V LDO Note: 28-5



Schematic design notice of "28_POWER_ThirdParty-Power"	
Note 28-1:	VA12 Layout placement please close to AP
Note 28-2:	VS2 Buck Layout placement please close to PMIC MT6357
Note 28-3:	VCN33 LDO Layout placement please close to MT6631
Note 28-4:	1.MT6691OOP/A Buck Layout Placement please close to LP4X/LP4 2.MT6691ZXP/A Buck Layout Placement please close to LP4X 3.If DRAM Application is LPDDR4 , MT6691ZXP/A NC
Note 28-5:	U2810 LDO Layout Placement Please close to LPDDR4X/LPDDR4 VDD1 power ball
Note 28-6:	For EMI_VDD2_FB and EMI_VDDQ_FB, please follow MMD rule

Title	14_POWER_ThirdParty_Powers	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER: YL
Date:	Saturday, May 21, 2022	Sheet 14 of 46

LPDDR4X



Schematic design notice of "44_Memory_eMMC_LPDDR4X_LPDDR4"

Note 44-1: Please refer to power supply related page select output voltage properly for LP4X/LP4

Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ,

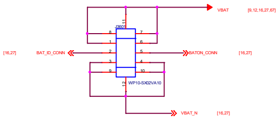
Note 44-3: Please refer to eMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of eMMC.

Note 44-4: VDD2 VDDQ decoupling cap: closed to DRAM ball.
For other cap for PMIC [>10uF, at PMIC page]:
please also refer to MMD and layout guide for placement.

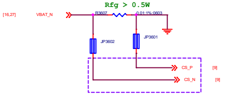
Note 44-5: Please check MT6765, MT6762 and MT6761's capacitor value.

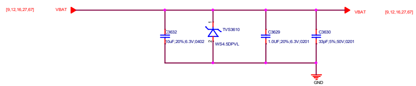
Title	15_Memory_eMMC_LPDDR4X		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	liufengjie
Date:	Saturday, May 21, 2022	Sheet 15 of 84	

Battery Connector



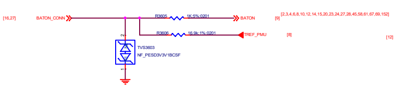
Guage Resistor





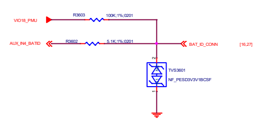
VBUS OVP

Battery NTC

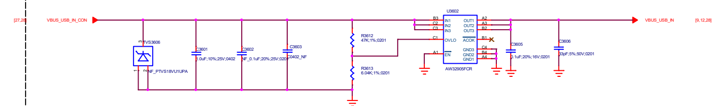


Battery pack is considered not present if BAT_ON is above 0.944*TREF

Battery ID



ADC 0.05-1.45 with 0.3 gap



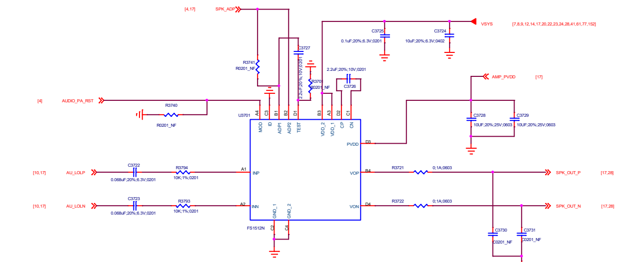
Surge VBUS: ±100V with Overvoltage Protection up to 29V DC

Vovp=10.5V when R3612=47K R3613=6.04K

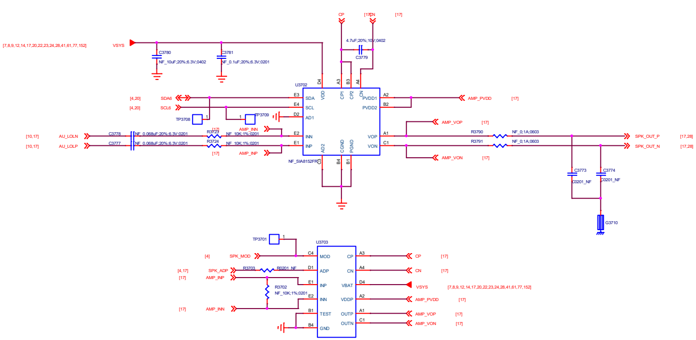
Vovp=1.2*(1+ R3612/ R3613)

Title	36_Battery/USB IF		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet	16 of 46

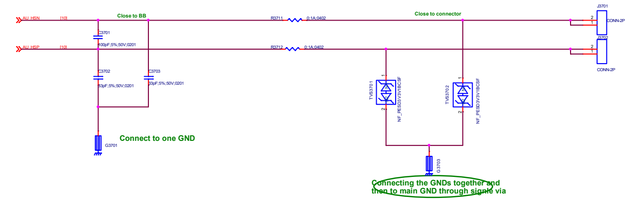
Audio PA-Selection



For FS1512N, R3741=0ohm,C3727=NC,R3701=0ohm
For SIA8102A & AW87318L, R3741=NC, C3727=2.2uF, R3701=NC.

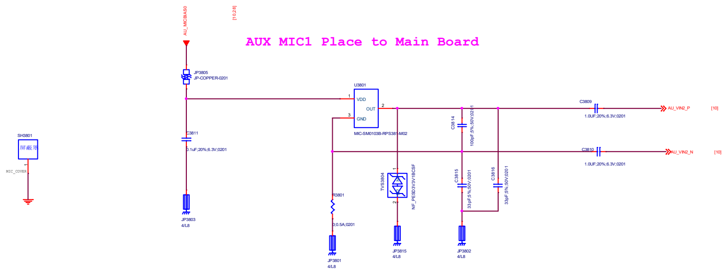


Receiver

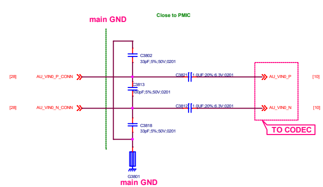


Title	37_SPK/Receiver/Vibrator	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
WINGTECH			
Date:	Saturday, May 21, 2022	Sheet 17	of 46

AUX MIC

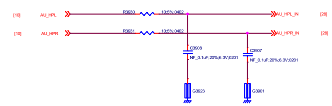


MAIN MIC

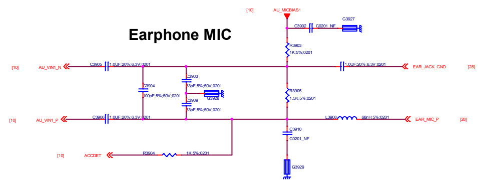


Title	38_MIC	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet	18 of 46

Earphone L/R



Earphone MIC



Title	39_Earphone	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT: WINGTECH-SH		DESIGNER: YL
WINGTECH		
Date: Saturday, May 21, 2022		Sheet 19 of 46

Note 62-1: Part # of BEAD6202, BEAD6203, BEAD6204 and BEAD6205 needs changed to "BLM18BD102SN1" for high THD performance (-90dB) but this BOM change will results in FM RSSI 10dB degraded .

Note 62-2: Reserved Cap for CS/RS test, please double check multi-key function when used

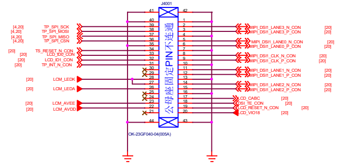
Note 62-3:

	Earphone Jack @ Main board	Earphone Jack @ Sub board
R6513	0 ohm	100nF

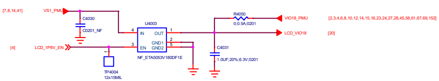
Note 62-4: Please Select ACC Mode for Operator Project to Pass Electrical MOS Test; More Information refer to Audio/Speech Design Notice

Note 62-5: Please select R6231 with 0402 size

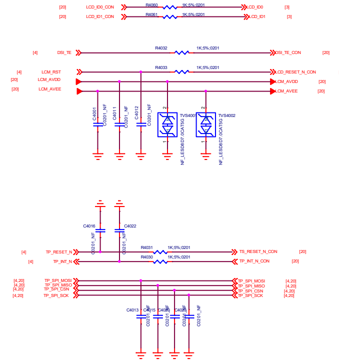
LCM BTB



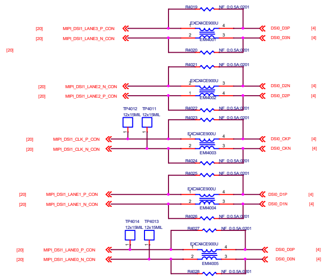
Reserve for ESD



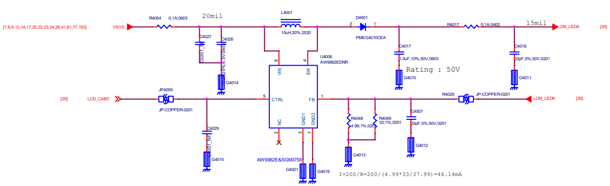
LCD Backlight LED Driver



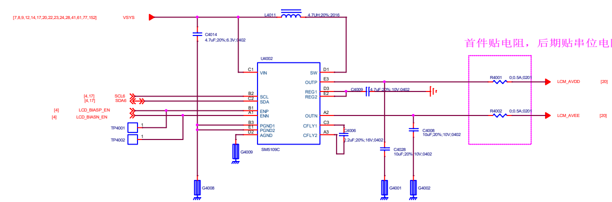
Common Mode Filter



LCD Backlight LED Driver

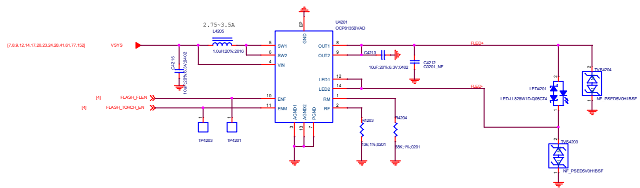


LCD Gate Drive



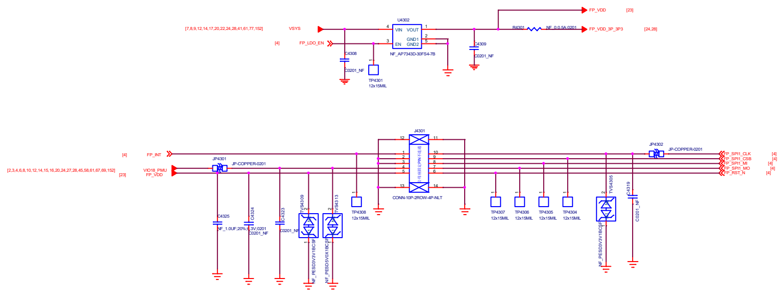
Title	40_LCD/CTP IF		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
WINGTECH			
Date:	Saturday, May 21, 2022	Sheet	20 of 46

Flash driver



Title	42_Flash/RGB	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER: YL	
Date:	Saturday, May 21, 2022	Sheet 22 of 46	

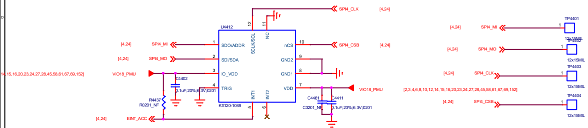
Fingerprint



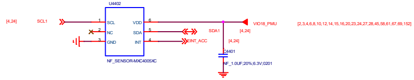
Title	43.Fingerprint	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
WINGTECH			
Date:	Saturday, May 21, 2022	Sheet	23 of 46

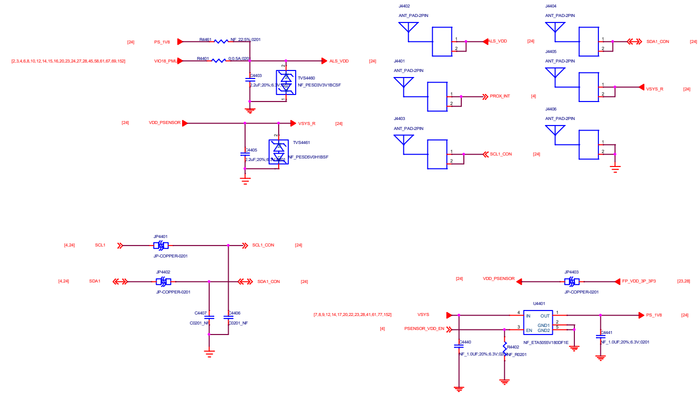
ALS/PROXIMITY SENSOR

A Sensor



兼容热式加速度计U4402

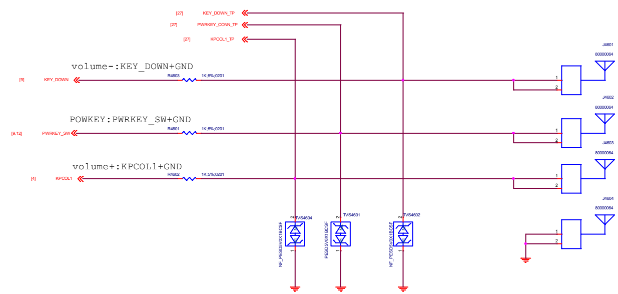




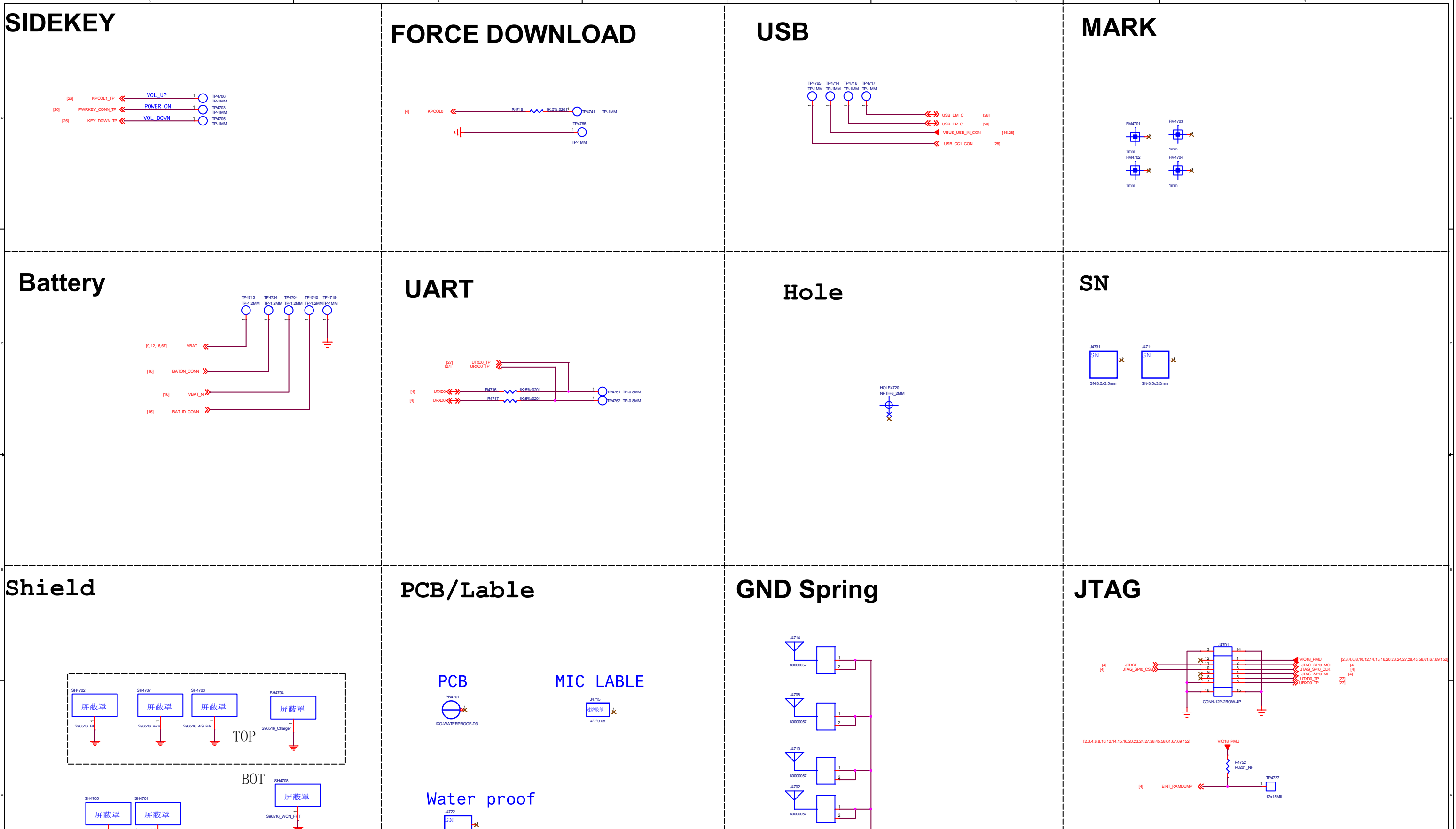
Title	44_Sensor	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet 24	of 52

Power Key / Key Pad

DO NOT put pull-up resistor on PWRKEY

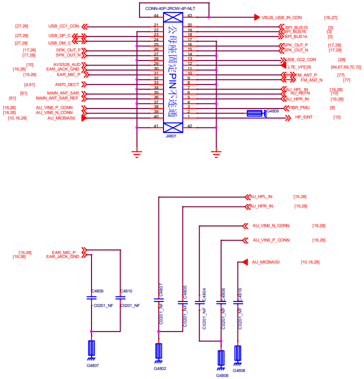


Title	46_Sidekey	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet 26	of 46

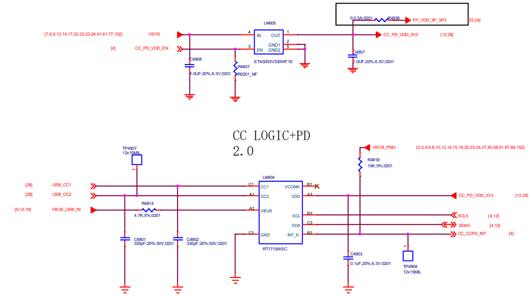


Title	47_Testpoint/Shielding/GND	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT: WINGTECH-SH		DESIGNER: YL
Date: Saturday, May 21, 2022		Sheet 27 of 46

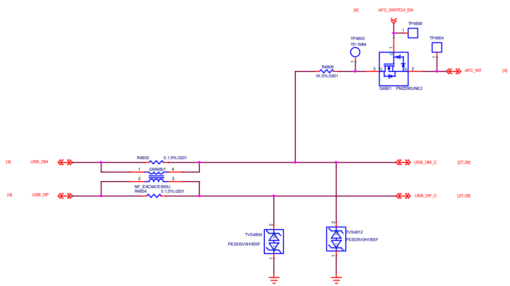
SUB CONNECTOR



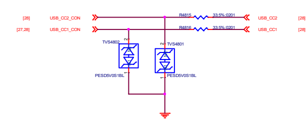
USB PD



USB_DP DM



USB CC

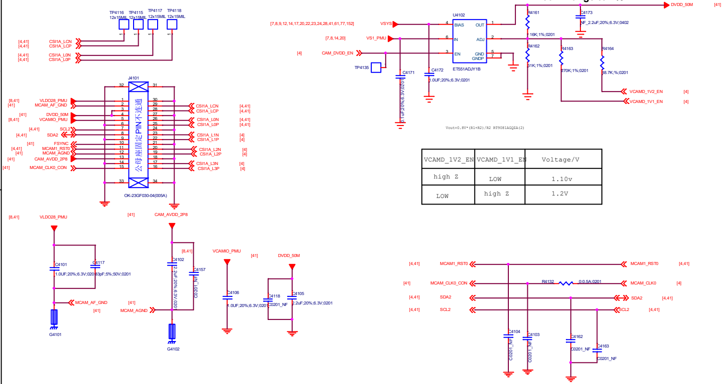


Title	48_Sub PCB IF	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER: YL
Date:	Saturday, May 21, 2022	Sheet 28 of 45

WINGTECH

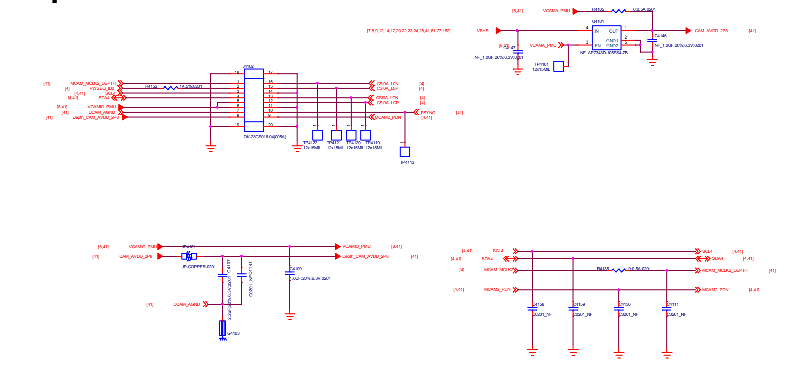
Main Camera 50M

R7502Change to 1%

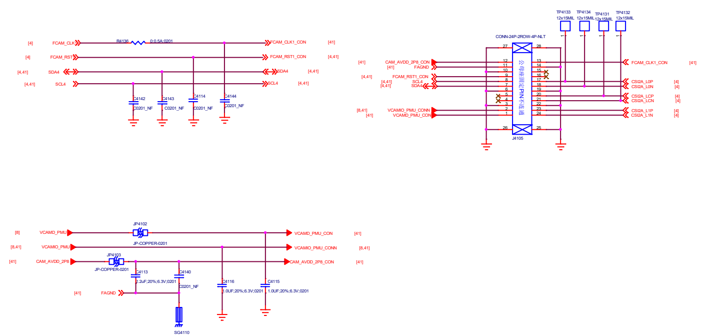


VCAMD_1V2_EN	VCAMD_1V1_EN	Voltage/V
high Z	LOW	1.10v
LOW	high Z	1.2V

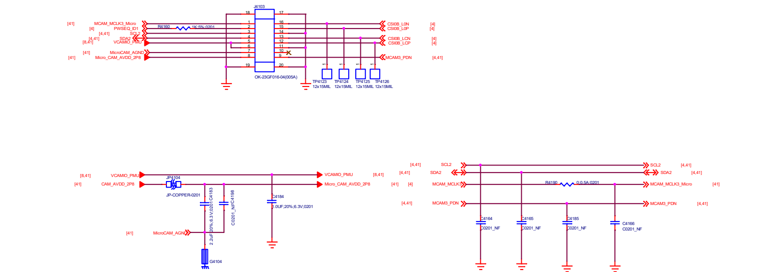
Depth Camera 2M



Front Camera 5M

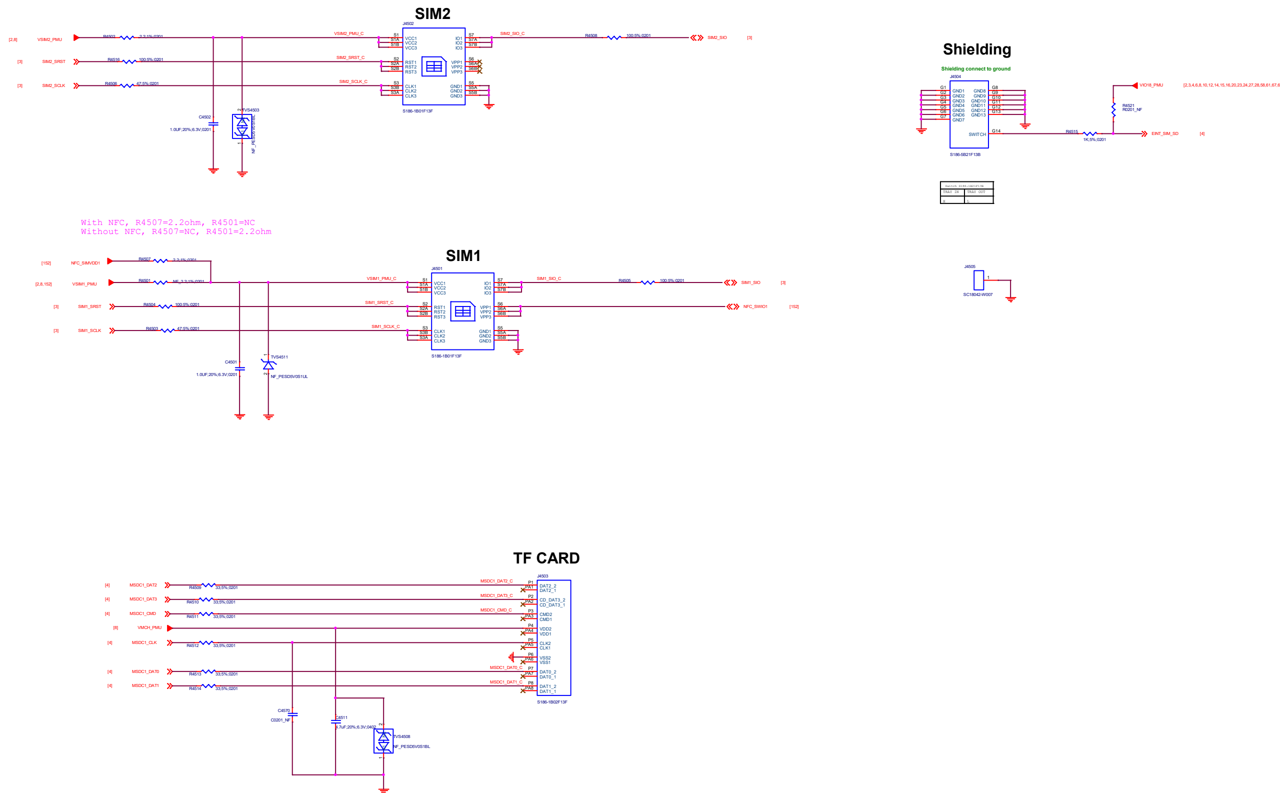


Micro Camera 2M



Title	41_Camera IF	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER: liufenglei
Date:	Saturday, May 21, 2022	Sheet 41 of 84

WINGTECH



Title	45_SIM/TF IF	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	liufenglei
Date:	Saturday, May 21, 2022	Sheet 45 of 84	

56. RF Interface

58. RF_MT6177_Pin_Out

61. Primary_ANT

62. Primary_ASM

64. TRX_LB

65. TRX_MHB

67. RF_MMPA_RF_TX

69. Diversity_ANT

70. Diversity ASM

72. DRX_LMB

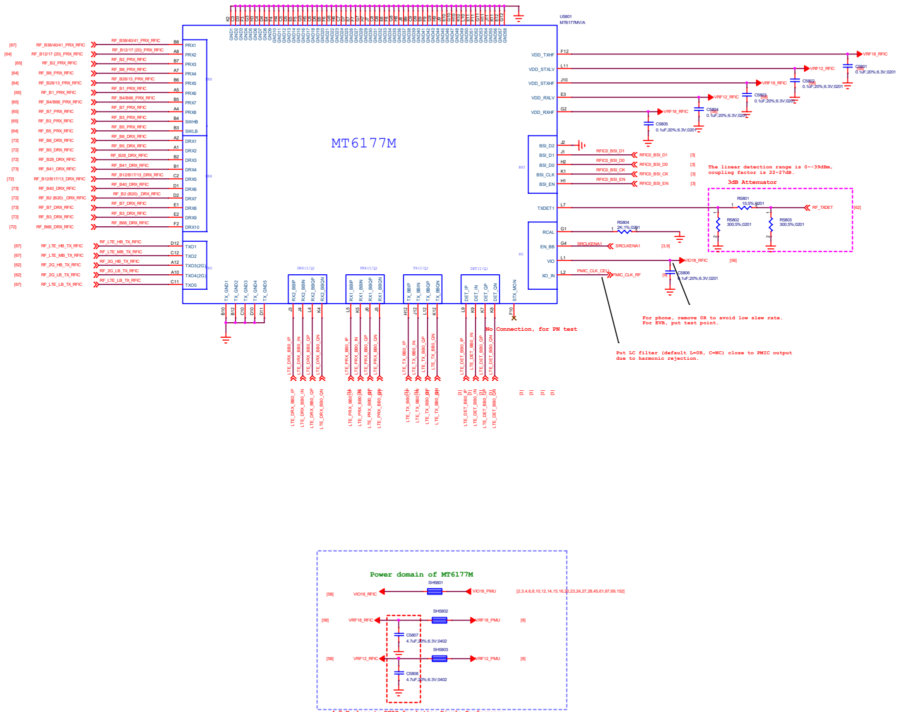
73. DRX_HB

77. WCN_MT6631

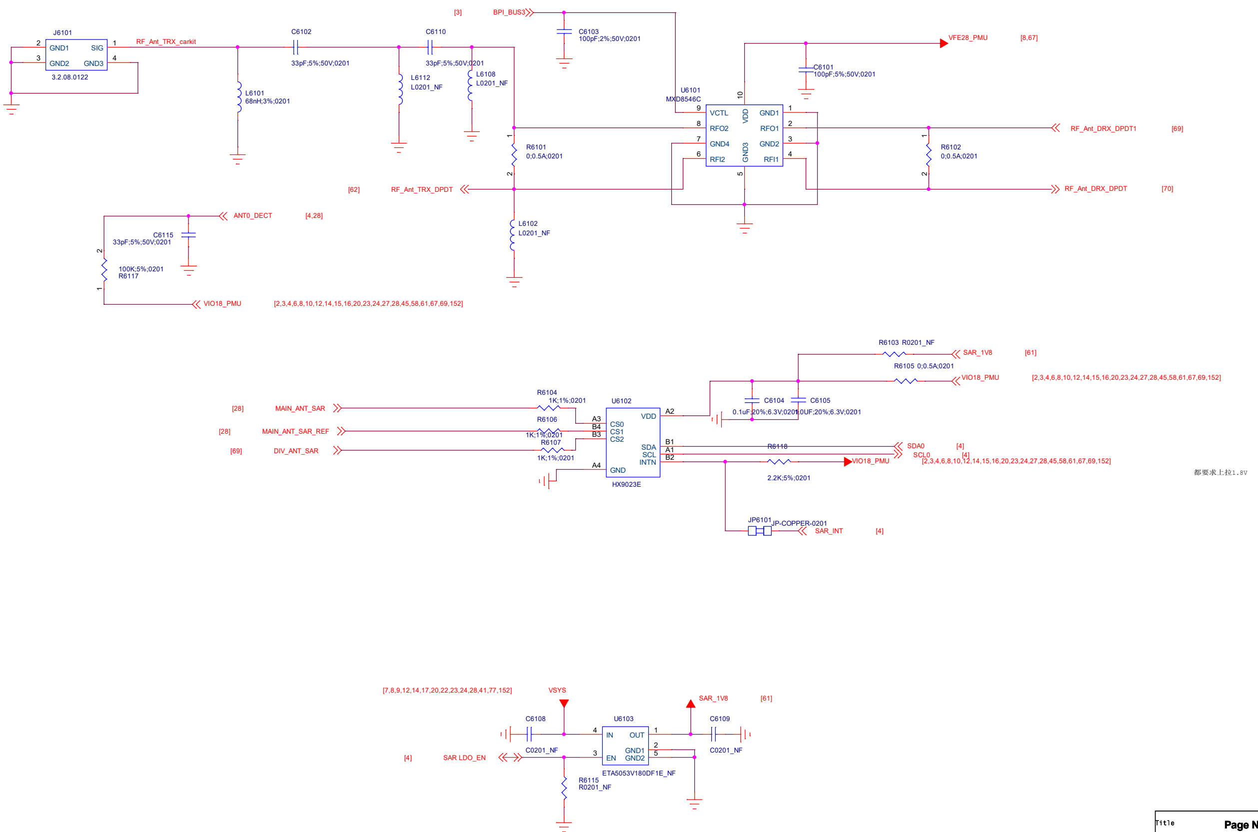
78. TCC303

79. NFC_PN553

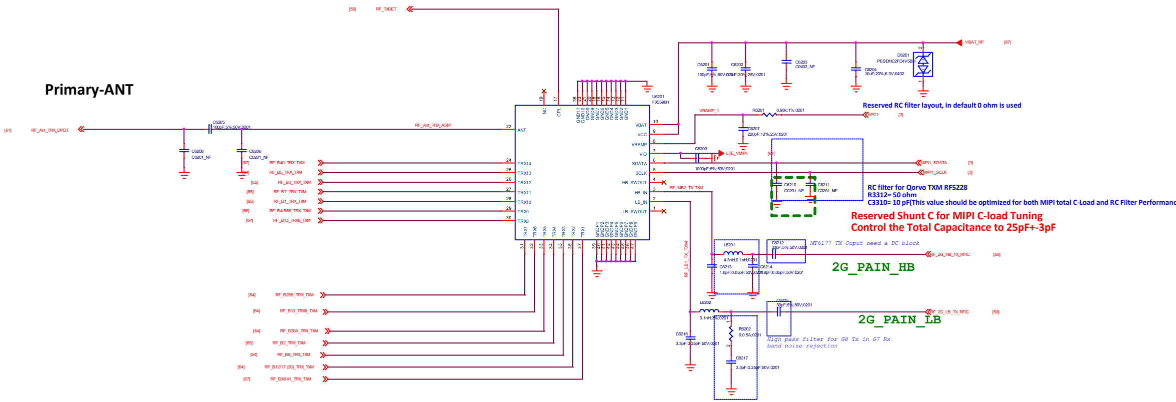
Title	Page Name = 56_RF Interface	REV: V10	
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523_[illegible]	D	
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Saturday, May 21, 202[illegible]	Sheet 56 of 81	



Title	Page Name = 58_RF_MT6177M	REV: V10
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523_1130	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Saturday, May 21, 2022	Sheet 58 of 81

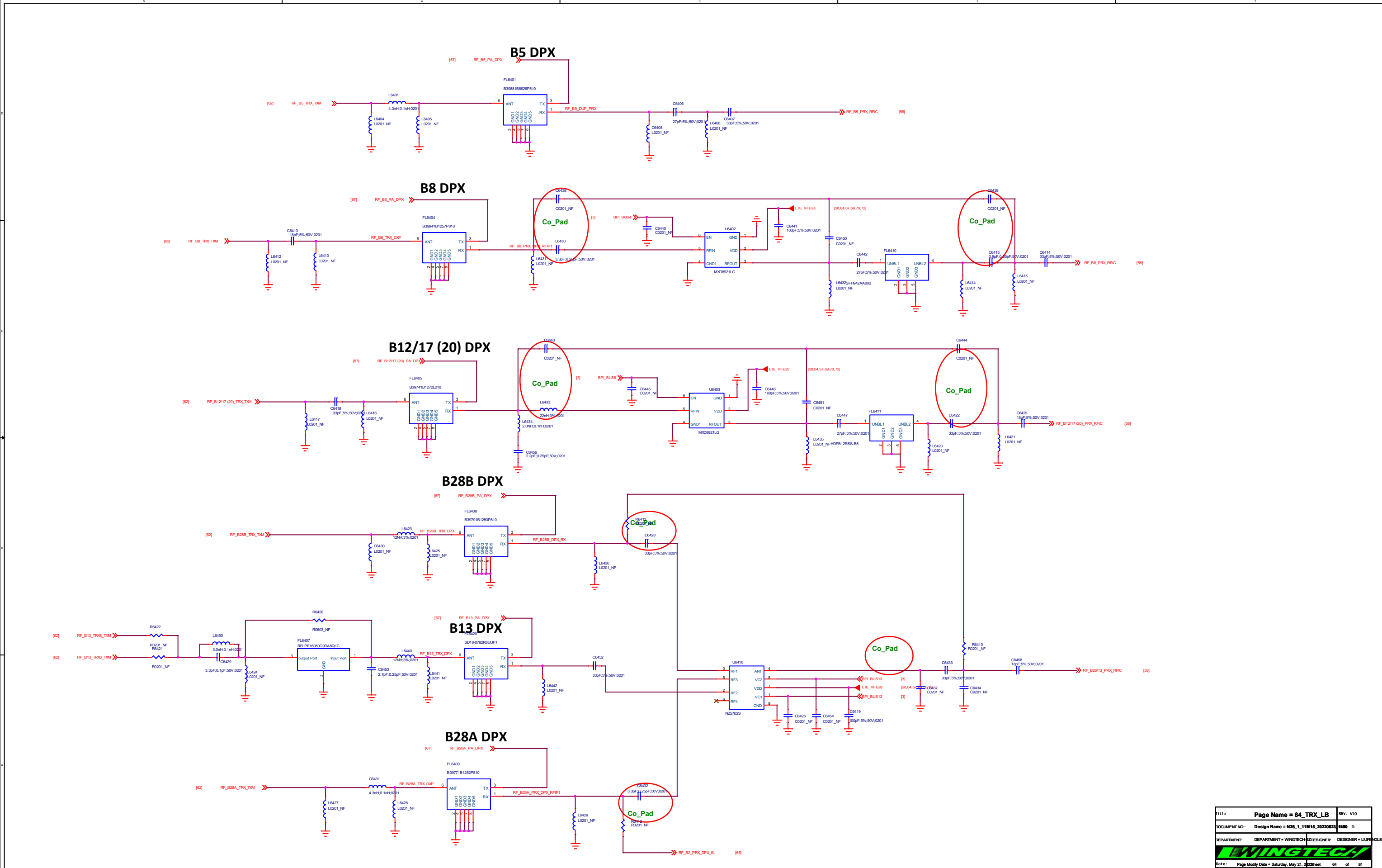


Title	Page Name = 61_Primary_ANT	REV: V10
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523_1136	
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLE
WINGTECH		
Date:	Page Modify Date = Saturday, May 21, 2022	Sheet 61 of 81

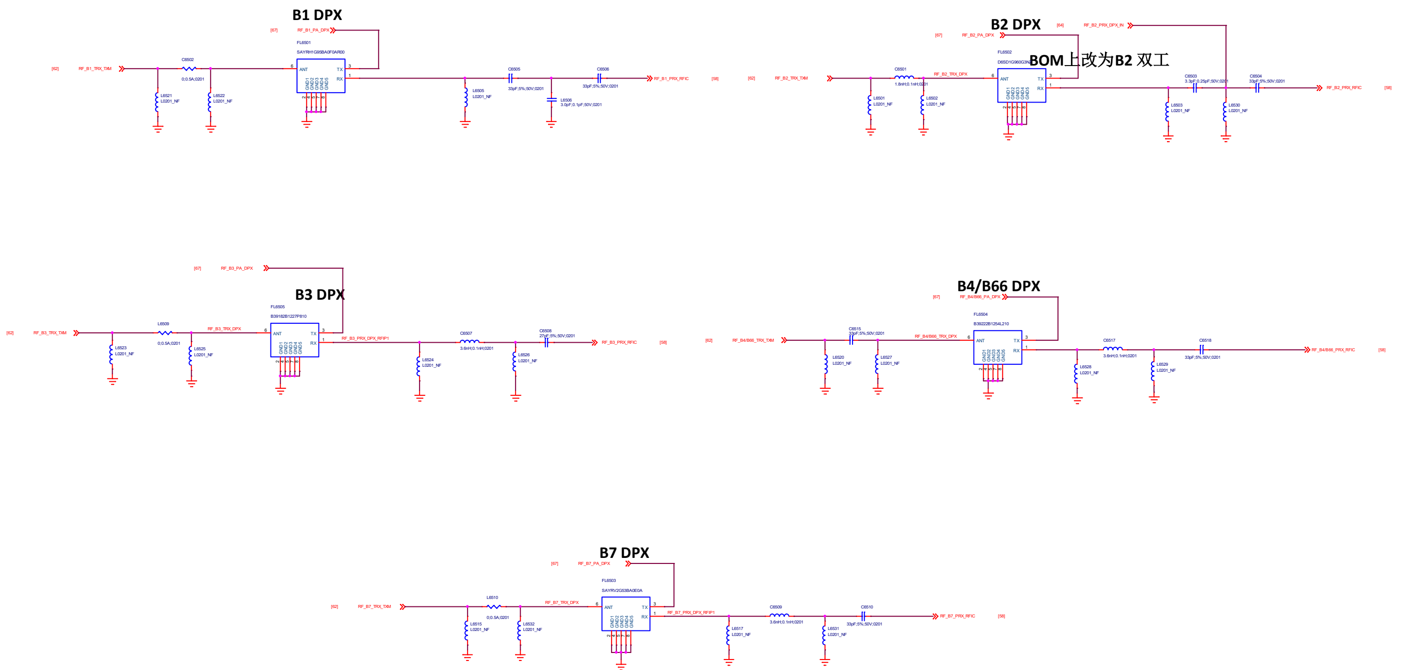


Title	Page Name = 62_Primary_ASM
Rev:	V10
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523
DEPARTMENT:	DEPARTMENT = WINGTECH
DESIGNER	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Saturday, May 21, 2022
Sheet	62 of 81

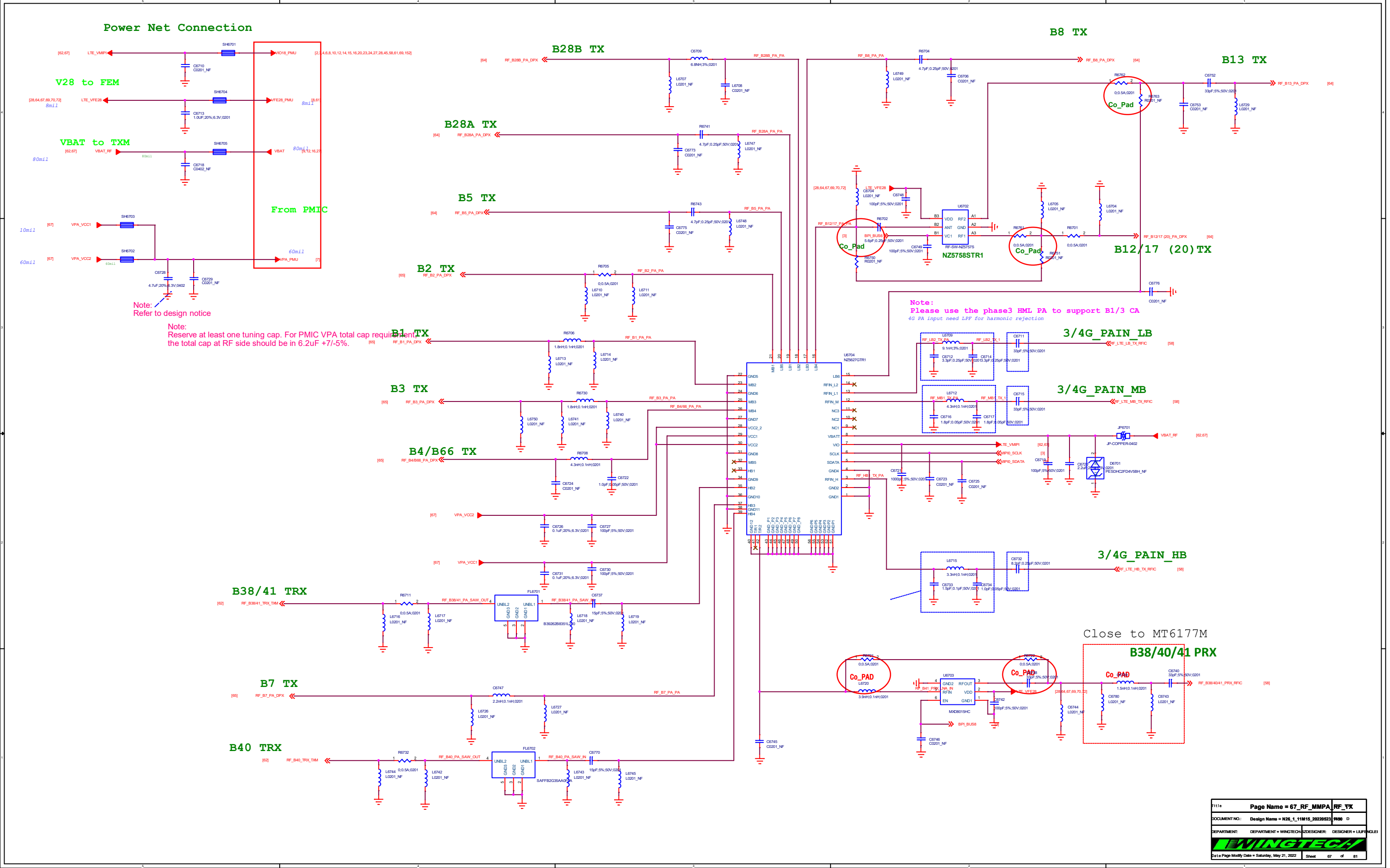
WINGTECH



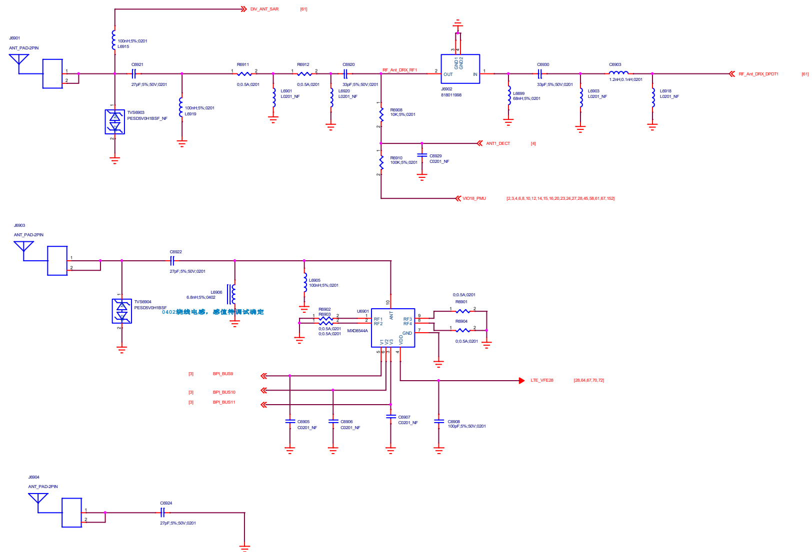
B5 DPX
RF_B5_PA_DPX
RF_B5_TRX_TXM
FL6401
B39881B8626P810
RF_B5_DUP_PRX
RF_B5_PRX_RFIC
B8 DPX
RF_B8_PA_DPX
RF_B8_TRX_TXM
FL6404
B39941B1257P810
RF_B8_TRX_DXP
Co_Pad
BPI_BUS4
LTE_VFE28
U6402
MXD8921LG
FL6410
RF_B8_PRX_RFIC
B12/17 (20) DPX
RF_B12/17 (20)_PA_DPX
RF_B12/17 (20)_TRX_TXM
FL6405
B39741B1272L210
BPI_BUS5
U6403
MXD8921LG
FL6411
RF_B12/17 (20)_PRX_RFIC
B28B DPX
RF_B28B_PA_DPX
RF_B28B_TRX_TXM
FL6408
B39791B1253P810
RF_B28B_TRX_DPX
RF_B28B_DPX_RX
B13 DPX
RF_B13_TRXB_TXM
RF_B13_TRXB_TXM
FL6407
FL6420
RF_B13_PA_DPX
RF_B13_TRX_DPX
U6410
NZ5762S
BPI_BUS13
LTE_VFE28
BPI_BUS12
RF_B28/13_PRX_RFIC
B28A DPX
RF_B28A_PA_DPX
RF_B28A_TRX_TXM
FL6409
B39771B1252P810
RF_B28A_TRX_DXP
RF_B28A_PRX_DPX_RFIP1
RF_B2_PRX_DPX_IN
Page Name = 64_TRX_LB
REV: V10
DEPARTMENT = WINGTECH
WINGTECH
Page Modify Date = Saturday, May 21, 2022
Sheet 64 of 81



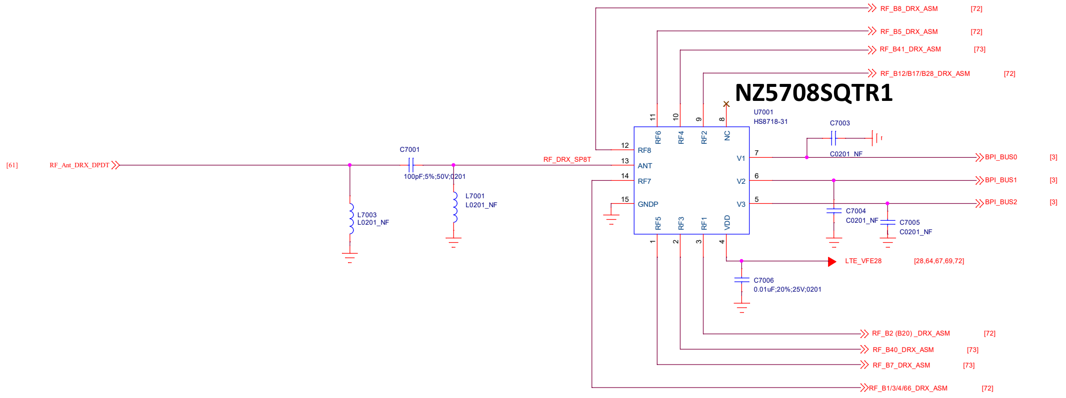
Title	Page Name = 65_TRX_MHB	REV: V10
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523_1180	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Saturday, May 21, 2022	Sheet 65 of 81



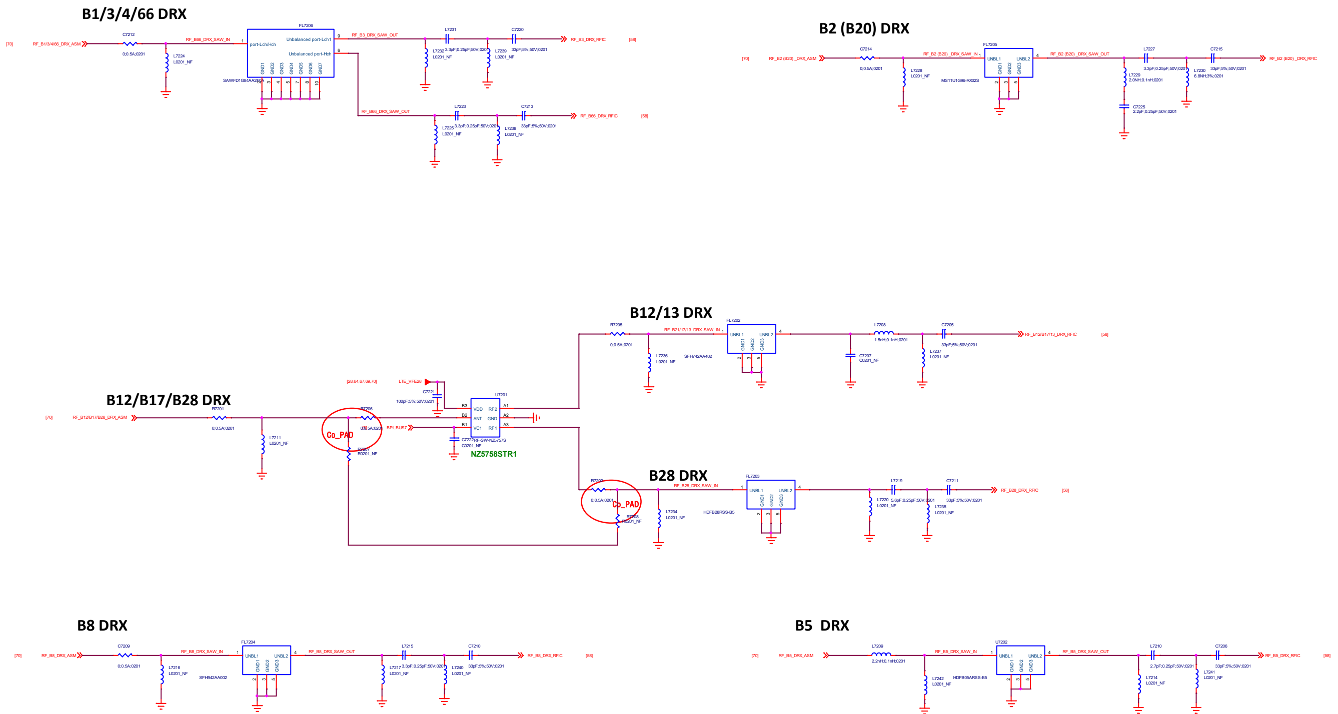
Power Net Connection
V28 to FEM
VBAT to TXM
From PMIC
Note:
Refer to design notice
Note:
Reserve at least one tuning cap. For PMIC VPA total cap requirement,
the total cap at RF side should be in 6.2uF +7/-5%.
B28B TX
B28A TX
B5 TX
B2 TX
B1 TX
B3 TX
B4/B66 TX
B38/41 TRX
B7 TX
B40 TRX
B8 TX
B13 TX
B12/17 (20)TX
Co_Pad
NZ5758STR1
Note:
Please use the phase3 HML PA to support B1/3 CA
4G PA input need LPF for harmonic rejection
3/4G_PAIN_LB
3/4G_PAIN_MB
3/4G_PAIN_HB
Close to MT6177M
B38/40/41 PRX
Co_PAD
Page Name = 67_RF_MMPA_RF_TX
Design Name = N26_1_11M15_20220523
DEPARTMENT = WINGTECH-SZ
DESIGNER = LIUFENGLEI
WINGTECH
Date Page Modify Date = Saturday, May 21, 2022
Sheet 67 of 81



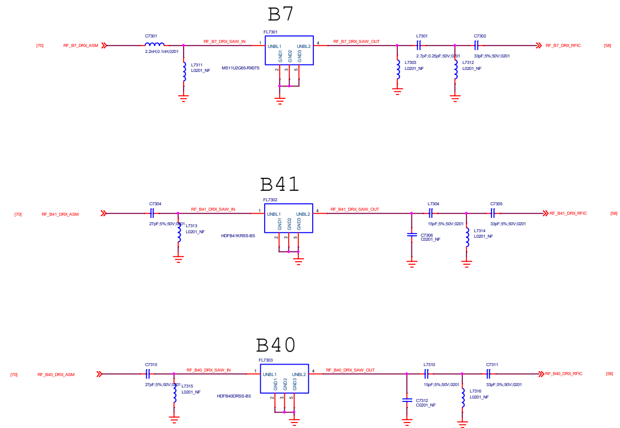
Title	Page Name = 69_Diversity_ANT	REV:	V10
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523_1930		D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Saturday, May 21, 2022	Sheet 69	of 81



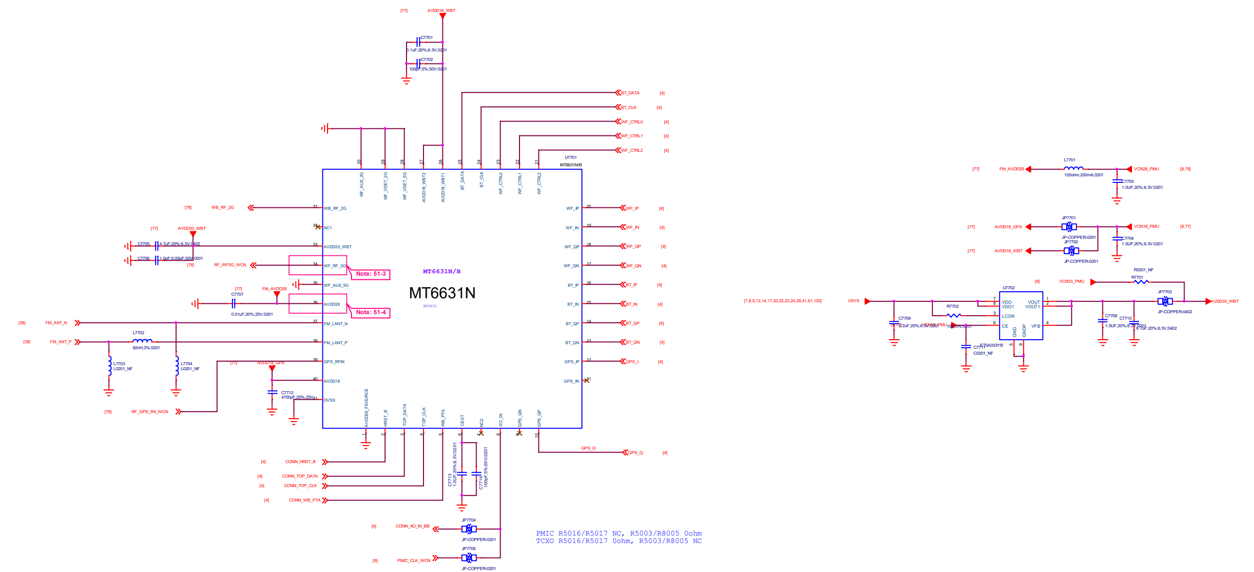
Title	Page Name = 70_Diversity ASM	REV: V10	
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523_1136	Size C	
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Saturday, May 21, 2022	Sheet 70	of 81



Title	Page Name = 72_DRX_LMB	REV: V10
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523_1930	D
DEPARTMENT:	DEPARTMENT = WINGTECH	DESIGNER: DESIGNER = LIUFENGLEI
WINGTECH		
Date:	Page Modify Date = Saturday, May 21, 2022	Sheet 72 of 81



Title	Page Name = 73_DRX_HB	REV: V10
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523_1930	D
DEPARTMENT: DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Saturday, May 21, 2022	Sheet 73 of 81



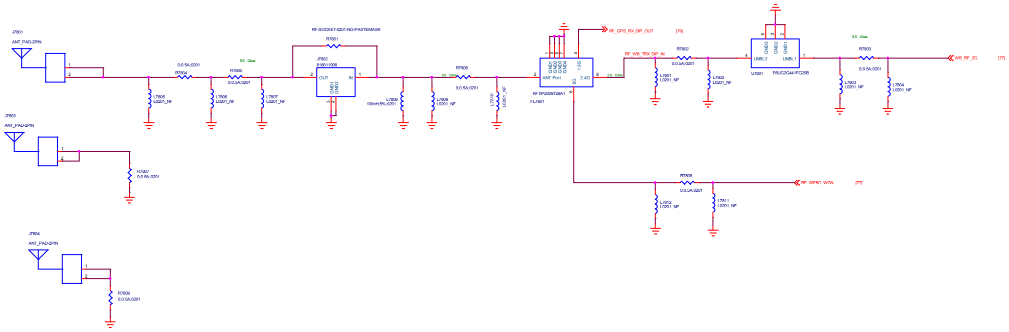
Schematic design notice of "51_CONNECTIVITY_CONSYS_MT6631"

Note 51-1:	For R5015 size, please select 0402 size or larger one
Note 51-2:	Please refer to MT6762 Baseband design notice for VCN33 LDO selection guide
Note 51-3:	If WiFi 5G not support, connect pin 34(WF_RF_5G) to GND
Note 51-4:	Pin 36 (AVDD28_FM) must be connected to VCN28 even if FM not support
Note 51-5:	If WiFi 5G were no need, VCN33 could be chosen from PMIC output (VCN33_PMU)

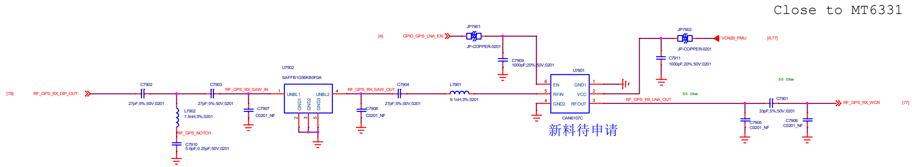
Title	Page Name = 77_WCN_MT6631
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523_1930
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ
DESIGNER:	DESIGNER = PENGLIZHONG
Date:	Page Modify Date = Saturday, May 21, 2022
Sheet	77 of 83

WINGTECH

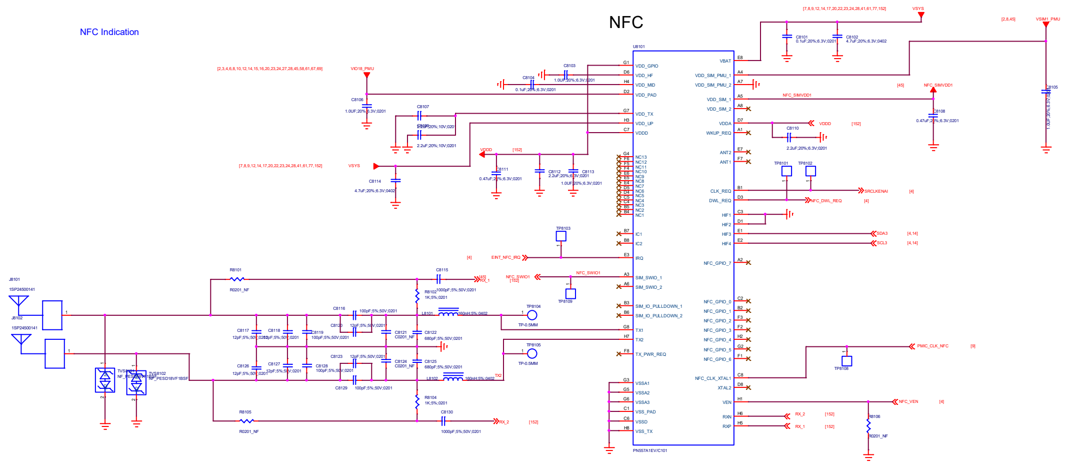
2.4G Wi-Fi BT GPS



Title	Reserved	REV: V10	
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	PENGLIZHONG
WINGTECH			
Date:	Saturday, May 21, 2022	Sheet 78	of 83



Title	Reserved		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	PENGLIZHONG
WINGTECH			
Date:	Saturday, May 21, 2022	Sheet 79 of 83	



Title	81_NFC_PN557	REV:	V10
DOCUMENT NO.:	[illegible]	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	liufenglei
Date:	Saturday, May 21, 2022	Sheet	152 of 152